

Tensor Omics: Core Preprocessing and Analysis Pipeline

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Overview

This document outlines the sequential pipeline used in Tensor Omics for preprocessing, vector space construction, outlier detection, and geometric analysis of gene expression data.

1. Normalization of Expression Values

Raw gene expression data is normalized in the following steps:

1. Division by gene-wise standard deviation for scale normalization.
2. Quantile normalization across samples to reduce global expression biases.
3. Log-transformation: $x \mapsto \log(1 + x)$ to stabilize variance.
4. For stress response studies, log fold-changes are computed:

$$\Delta_{\text{stress}} = \log(1 + x_{\text{stress}}) - \log(1 + x_{\text{control}})$$

2. Construction of Expression Vector Space

- Biological replicates belonging to the same axis (tissue, stressed tissue, etc.) are averaged post-normalization.
- Each gene is represented as a vector $\vec{p} \in \mathbb{R}^n$, where n is the number of conditions or tissues. Thus n is the number of dimensions of the vector space.
- Expression vectors are stored in Fortran arrays. *Importantly* Fortran is column major and vectors *must* be columns.
- Two K-D Trees are constructed:
 - A raw-space K-D Tree (unnormalized vectors) for Euclidean queries.
 - A normalized-space K-D Tree (unit-length vectors) for cosine similarity and angular queries.
- The first axis is designated the **Anchor Axis (AA)** for visualization.

3. Gene Family Grouping and Centroid Estimation

- Gene families, populations, or functional groups are defined using metadata.
- Family centroids $\vec{o}$ are computed either as:
 - The mean vector of all family members.
 - The mean vector of a family’s orthologs only.
- All metadata and vectors can be serialized into `.txdata` binary files.

4. Empirical Distribution Estimation

To detect expression outliers and classify divergence types, we estimate empirical distributions for four key geometric indices across gene families or defined subsets:

- **Relative Distance Index (RDI)**: For each gene vector $\vec{p}$, its Euclidean distance to the family centroid $\vec{o}$ is computed as:

$$\text{RDI} = \|\vec{p} - \vec{o}\|$$

All RDI values across families are collected and percentile thresholds (e.g. 95%) are used to identify significant divergence in magnitude.

- **Relative Angle Index (RAI)**: The angular difference between a vector $\vec{p}$ and its centroid $\vec{o}$ is computed using the cosine formula:

$$\text{RAI} = \cos^{-1} \left(\frac{\vec{p} \cdot \vec{o}}{\|\vec{p}\| \cdot \|\vec{o}\|} \right)$$

This identifies divergence in direction. Thresholds (e.g. 95%) are derived from the pooled distribution of angles.

- **Relative Axis Signal (RAS)**: For each unit-normalized expression vector $\vec{p}^{\text{norm}}$, we consider its component $\vec{p}_i^{\text{norm}}$ along each axis i (e.g., tissue). For each axis, the 5th percentile across all vectors is computed as the inactivity threshold. A gene is considered neofunctionalized in axis i if:

$$\vec{o}_i^{\text{norm}} < T_i^{(5\%)} \quad \text{and} \quad \vec{p}_i^{\text{norm}} > T_i^{(5\%)}$$

where $\vec{o}^{\text{norm}}$ is the centroid.

- **Relative Frobenius Index (RFI)**: For any set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ assembled as columns in matrix $M \in \mathbb{R}^{d \times n}$, the signal strength is computed as:

$$\text{RFI} = \frac{\|M\|_F}{\sqrt{n}} = \frac{\sqrt{\sum_{i=1}^d \sum_{j=1}^n M_{ij}^2}}{\sqrt{n}}$$

This provides a normalized estimate of signal per vector and is used to assess subfield strength.

All empirical distributions are stored in Fortran arrays and used to assign significance thresholds for downstream classification and visualization.

5. Outlier Detection and Shift Vector Calculation

- Outliers are detected when $d_{po} = \|\vec{p} - \vec{o}\|$ exceeds the 95th percentile RDI. *Importantly* the distance d_{po}^{norm} is greater or equal than the 95 percentile of RDI distances. So, for outlier detection we use the normalized distance, divided by the max family distance, but for all later considerations we use the original, not normalized, distance.
- Next, shift vectors are computed as:

$$\vec{s} = \vec{p} - \vec{o}$$

- Both shift vectors and outlier indices are stored for further analysis.

6. Tissue Versatility Calculation

- Cosine of angle between expression vector and space diagonal:

$$\cos(\theta) = \frac{\vec{p} \cdot \vec{d}}{\|\vec{p}\| \cdot \|\vec{d}\|}$$

where $\vec{d} = (1, 1, \dots, 1)$ is the space diagonal.

7. Clock Plot Projection and Rotation Angle Computation

- All shift vectors are projected into the **Relative Axes Plane (RAP)**, orthogonal to the space diagonal and intersecting the origin.
- The following quantities are calculated for each outlier:
 - Projected shift vector in RAP.
 - Normalized (unit-length) projection.
 - Clock hand vector from origin to projected point.
 - Rotation angle (standardized as clockwise) between projected centroid and outlier.
- Results are stored in a structured Fortran matrix:
 - For each outlier i , store:
 - * Rows 1 to n : $\vec{s}_i^{\text{RAP}}$
 - * Rows $n + 1$ to $2n$: $\vec{s}^{\text{RAP, norm}}$
 - * Rows $2n + 1$ to $3n$: $\vec{s}_i^{\text{clock}}$
 - * Row $3n + 1$: rotation angle in radians